

07 Jul 2026 05:22:57 ET | 16 pages

# BOE Technology (000725.SZ)

## Downgrade to Neutral as Glass Substrate Expectations Likely Priced In

### CITI'S TAKE

We attended BOE's investor day. BOE reiterated its position as the global leader in displays while emphasizing that its next phase of growth will be driven by the extension of its core capabilities into glass-based advanced packaging, optical interconnect, perovskite photovoltaics, foldable glass, etc. BOE anticipates reaching a definitive mass-production investment decision by mid-2027. Its optical interconnect devices remain at a validation stage and have not yet entered mass production, but BOE targets to launch mature multi-channel product demos aligned with 1.6T, 3.2T and 6.4T industry requirements in 2028. We downgrade BOE from Buy to Neutral with a new TP of Rmb8.7 (prev: Rmb5.0), as we believe current valuation has factored in an improving LCD biz and potential glass substrate contribution.

**BOE's second and third growth curves are increasingly material** — Since launching its IoT transformation in 2021, BOE has achieved global No.1 shipment positions in more than 20 sub-segments, including automotive display, commercial display and public healthcare. Innovation business revenue is expected to reach Rmb60bn in 2026, representing around 30% of total revenue and becoming a major growth pillar, according to mgmt. The third-curve incubation portfolio includes perovskite photovoltaics, glass-based packaging substrates, optical communication / interconnect, automotive dimming glass, regenerative medicine, robotics, VR/AR and 3D light-field display.

**Cooperation with Corning** — BOE and Corning have forged a strategic partnership to unlock global market opportunities across four key technological frontiers. In glass-based packaging substrates, BOE is combining Corning's TGV formulations with its experience in large-scale manufacturing to execute process qualification, technology transfer, and domestic market expansion. In optical interconnect, BOE will combine its micro-LED technology with Corning's optical systems to achieve mass-production readiness for next-generation AI data transmission. They are also validating foldable and bendable glass to expand beyond smartphones into broader automotive & IT applications. In perovskite photovoltaics, BOE is testing Corning's specialized UV-converting glass to enhance module efficiency & lifespan. Propelled by these joint validations, BOE targets a definitive mass-production investment decision for the perovskite business by late 2026 or early 2027, anticipating it will scale into a major business pillar by 2028–29. *(continued inside/below...)*

### Earnings Summary

| Year to | Net Profit | Diluted EPS | EPS growth | P/E  | P/B | ROE | Yield |
|---------|------------|-------------|------------|------|-----|-----|-------|
| 31 Dec  | (RmbM)     | (Rmb)       | (%)        | (x)  | (x) | (%) | (%)   |
| 2024A   | 5,323      | 0.141       | 110.5      | 54.9 | 2.2 | 4.1 | 0.6   |
| 2025A   | 5,857      | 0.156       | 10.4       | 49.7 | 2.2 | 4.4 | 0.7   |
| 2026E   | 7,560      | 0.203       | 30.1       | 38.2 | 2.0 | 5.5 | 0.9   |
| 2027E   | 9,469      | 0.256       | 25.9       | 30.4 | 1.9 | 6.5 | 1.2   |
| 2028E   | 12,037     | 0.325       | 27.1       | 23.9 | 1.8 | 7.6 | 1.5   |

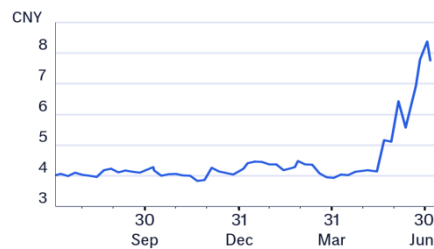
Source: Powered by dataCentral

### Neutral ↓ from Buy

|                             |               |
|-----------------------------|---------------|
| Price (06 Jul 26 15:00)     | Rmb7.760      |
| Target price                | Rmb8.700↑     |
|                             | from Rmb5.000 |
| Expected share price return | 12.1%         |
| Expected dividend yield     | 0.9%          |
| Expected total return       | 13.0%         |
| Market Cap                  | Rmb285,023M   |
|                             | US\$42,105M   |

### Price Performance

(RIC: 000725.SZ, BB: 000725 CH)



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See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

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**Glass-based packaging substrates.** While glass substrates are emerging as superior alternatives to traditional PCBs for high-end AI and HPC applications, BOE is rapidly advancing its industrialization. Leveraging over six years of experience, BOE established mainland China's first glass substrate pilot line in 2024 and successfully delivered second-generation samples in 2025. The company has overcome major technical bottlenecks, achieving significant breakthroughs in TGV drilling (reaching a 20:1 aspect ratio), void-free metal filling, high-density routing (supporting over 20 build-up layers with sub-2 $\mu$ m line widths), and chip-free cutting. Crucially, BOE's substrates have passed rigorous industry reliability tests, demonstrating very low warpage even during high-temperature reflow soldering. Over the next two years, BOE aims to prioritize yield optimization, stable mass-production capabilities, and innovative layer-reduction designs alongside ecosystem partners. BOE anticipates reaching a definitive mass-production investment decision by mid-2027.

**Optical interconnect and micro-LED light source.** BOE is advancing its optical interconnect technologies, highlighted by its development of a micro-LED light source featuring a 1.8GHz response speed and a 3Gbps transmission rate. BOE is also working with Corning and other partners on high-purity optical fiber, optical bridges, waveguides and related optical transmission technologies. To drive technology implementation, BOE has outlined a clear roadmap: it plans to launch a two-dimensional multi-channel demo by 2027, followed by mature product demos meeting 1.6T, 3.2T, and 6.4T industry requirements in 2028. Ultimately, BOE aims to transition these innovations into mass production, expanding their application to data centers and automotive interconnect scenarios.

## Cooperation with Corning

The disclosed core cooperation areas with Corning include glass-based advanced packaging substrates, optical interconnect, foldable or bendable glass, and perovskite glass materials. The parties aim to jointly develop global markets and potentially create one or multiple Rmb100bn-scale new business opportunities.

**The first core cooperation area is glass-based packaging substrates.** In the AI era, rising computing power is driving the need for larger IC sizes, higher packaging stack complexity and more stringent physical-layer requirements. Traditional PCB substrates increasingly face limitations in warpage, flatness and heat resistance. Corning will provide glass and process technologies specifically developed for through-glass via, or TGV, applications. BOE will contribute its strengths in glass-based processing and large-scale manufacturing. The cooperation will proceed in three steps: (1) Complete systematic qualification of the new glass formulation and related processes. (2) Technology transfer. (3) Joint development of the domestic strategic market for glass-based advanced packaging. For BOE, the cooperation priorities include introduction experiments for high-performance glass substrates, optimization of key processes such as total thickness variation, or TTV, and improvement of substrate reliability.

**The second core cooperation area is optical interconnect.** With AI driving significant data transmission demand, traditional copper interconnects face bandwidth, power consumption and heat dissipation bottlenecks. BOE and Corning believe the industry is moving from copper to optical transmission. Corning can provide optical fiber, connectors and optical system technologies, reflecting its full-stack capability in optical transmission. BOE will contribute its micro-LED light-source technology. BOE noted that related optical interconnect

technologies are also a core enabler for future Glass Bridge technology. The companies aim to jointly advance technology implementation and mass-production readiness.

**The third core cooperation area is foldable or bendable glass.** The initial focus is material validation and system integration. Future market expansion will not be limited to smartphones. BOE and Corning will focus on broader applications in automotive and IT, where foldable and bendable glass could unlock a larger addressable market.

**The fourth core cooperation area is perovskite photovoltaics.** Corning will provide specialty glass materials that can convert ultraviolet light into visible light. By leveraging the inherent properties of the glass itself, the material can improve perovskite conversion efficiency without a coating layer. Because there is no coating layer, the efficiency enhancement will not degrade over time, allowing the glass to meet the 15-20 year lifetime requirement of perovskite modules. For BOE, the collaboration with Corning will focus on ultraviolet performance optimization, power degradation improvement and photovoltaic module compatibility testing. BOE expects the perovskite business to meet the conditions for a mass-production investment decision by end-2026 or early 2027, subject to requirements on power-generation efficiency, stability and product lifetime. Management expects perovskite to gradually become an important business in 2028-29.

## Glass-based packaging substrates

High-end packaging substrates need to support large area with low warpage, high-density routing and low dielectric loss. Traditional PCB materials face growing limitations in AI and HPC applications. Glass substrates have three key advantages versus traditional materials: (1) Coefficient of thermal expansion, or CTE, close to silicon, resulting in low warpage. (2) Ability to achieve very small via diameters, supporting high-density packaging for AI chips. (3) Strong thermal performance, supporting heat dissipation in high-computing-power scenarios.

Packaging-related glass technologies can be divided into three categories (1) Panel-level packaging, or PLP. Square glass is used as a carrier and then peeled off in wafer-level packaging. (2) Glass interposer. Used for high-end, high-density direct die-to-die connection. (3) Glass substrate. Serves as the connection layer between the die and the PCB. Among these, glass substrate is currently the direction with the fastest industrialization progress.

The major technical bottlenecks include: (1) TGV drilling. The process must solve 20:1 aspect-ratio hole drilling, simultaneous processing of dense/sparse and large/small holes, roundness qualification, no open circuits and no hidden cracks. (2) Metal filling. Electroplating uniformity must be ensured. Surface copper and via copper thickness need to be consistent. Filling must be void-free, while stress between metal and glass must be managed. (3) Build-up and cutting. Multi-layer build-up can create stress accumulation and line breaks. Cutting brittle glass after build-up requires chip-free cutting technology. Automated inspection solutions must also be developed for transparent glass.

BOE has accumulated more than six years of TGV packaging process experience. In 2020, the company began work on single-process and material validation. In 2022, BOE invested in a 510mm × 515mm large-panel-width pilot line. The line achieved full process connection six months after equipment move-in. BOE also

has an 8-inch experimental line platform for basic material and formulation validation. In 2024, BOE built mainland China's first innovative pilot line for glass-based substrates. The company has completed customer conceptual validation. In 2025, BOE delivered second-generation samples. The 510mm × 515mm line is a complete mass-production validation line with investment of several hundred million Rmb, including more than Rmb100mn in automation equipment. It can validate automated yield and board-level small-sample production, providing high accuracy for future mass-production evaluation.

BOE has achieved meaningful process progress: (1) TGV process. Aspect ratio can reach 20:1. Waist diameter is adjustable between 65μm and 100μm. Roundness exceeds 90%. There are no cracks at the corner or via wall. (2) Filling process. Electroplating uniformity is below 5%. Surface copper and via copper thickness consistency reaches 80%. There are no blocked holes or bubbles. (3) Routing process. Line width is below 2μm. Line spacing is above 2μm. Alignment accuracy is 1μm. Routing uniformity is below 5%, with no broken lines. BOE can achieve more than 20 build-up layers and vertical stacked vias, supporting high-density interconnect requirements. (4) Cutting process. Chip-free cutting can be achieved, with 90-degree cutting angles that are 100% vertical.

BOE's single-layer substrate has passed industry-standard reliability tests, including 1,000 temperature cycles, HAST, high/low-temperature storage, and high-temperature/high-humidity aging. A 75mm × 75mm sample showed room-temperature warpage of only 40μm. It maintained low warpage after being heated to 260°C for reflow soldering, supporting AI and HPC chip requirements.

Over the next two years, BOE's key priorities are yield improvement and stable, sustainable mass-production capability. The company will also strengthen cooperation with ecosystem partners to overcome remaining process challenges and further explore the core value of glass substrates in layer-reduction design and packaging architecture optimization. Management emphasized that beyond large-area advantages, glass-based packaging substrates also create value through layer-reduction design. By optimizing packaging architecture using rectangular routing paths, glass substrates could potentially deliver meaningful system-level value. BOE expects to reach the conditions for a mass-production investment decision for glass substrates by mid-2027.

## Optical interconnect and micro-LED light source

BOE has established joint project teams with upstream and downstream enterprises and universities. Current progress includes a micro-LED light source with 1.8GHz response speed and 3Gbps transmission rate. Compared with VCSEL and other light sources, micro-LED has lower power consumption and lower temperature sensitivity. It can operate stably above 100°C, improving data stability while reducing temperature-control and heat-dissipation energy consumption. The company is working on high-speed photodiodes, optical drivers, electrical drivers, electrical chips and transmission protocols, with a goal of building a mature full-chain solution. BOE is also working with Corning and other partners on high-purity optical fiber, optical bridges, waveguides and related optical transmission technologies. Optical interconnect devices are still at a validation stage and have not yet entered mass production. BOE aims to continue to work with domestic and overseas research institutions and industry partners to advance technology implementation. BOE's roadmap is as follows (1) In 2027, to launch a two-dimensional multi-channel micro-LED optical interconnect solution demo. (2)

In 2028, to launch mature multi-channel product demos aligned with 1.6T, 3.2T and 6.4T industry requirements. (3) To expand applications into data centers, automotive interconnect and other scenarios.

### Third-generation semiconductor and GaN

BOE Huacan has completed construction of a gallium nitride (GaN) pilot line and now has preliminary mass-production conditions. Its 650V high-voltage product has passed customer validation. The company expects to launch a 40V low-voltage product by end-2026. It plans to later expand into automotive-grade and AI-scenario applications. In GaN power electronics, BOE Huacan has delivered the second high-voltage GaN device product to customers through its Yiwu GaN pilot line. GaN products remain at an optimization stage. Related products are likely to launch in 2027, with future applications targeting AI, industrial and automotive-grade power chips. Large-scale investment may be considered at the appropriate time, according to the company.

### Display

**D&A and capex.** Core production lines are generating attractive profitability, according to the company, with the Chongqing and Fujian Gen-8.5 lines both recording net margins above 20%. The two Gen-10.5 lines have continued to increase shipment volume through quality and efficiency improvements without capacity expansion. Depreciation likely peaked in 2025; although there may be a modest increase in 2027 due to depreciation from B16, BOE said the overall depreciation trend is downward. It also expects capex to decline after completion of the OLED Gen-8.6 line.

**LCD TV industry consolidation is likely close to completion.** The concentration ratio of the five largest LCD TV panel makers has remained flat in 2026 versus 2025, and BOE expects leading display companies to capture a larger share as industry concentration rises. It believes rising memory prices have a greater impact on smaller-size TV products, pushing the market towards larger-size TVs. The shipment share of 70–75-inch TVs increased from 8% in 2025 to 22% in 2026. BOE expects shipments of 80-inch and above TVs to increase by more than 20% in 2026 versus 2025. This should drive a continued increase in average TV size. BOE expects the average TV shipment size to reach 55 inches in 2027. Management views 55 inches as a key threshold for the TV panel industry. Once the average TV shipment size reaches 55 inches, it believes the TV segment could reach supply-demand balance. BOE expects this to happen in 2027, which would be favorable for BOE as a leader in large-size panels.

**Competition in the OLED industry could persist over the next two years.** BOE expects global OLED smartphone demand in 2026 to decline by 10–17%, mainly due to memory price increases pushing up terminal prices and suppressing consumer demand. Domestic smartphone brands are under pressure. Despite this, BOE's OLED business maintained stable growth in 1H26. The business, however, remains loss-making at the operating level. Management expects competition in the OLED industry to persist over the next two years. Supply volume to key customers continued to grow in 1H26.

**Memory price impact.** Downstream terminal brands are responding to high memory prices in two main ways, according to BOE. First, they are reducing specifications to alleviate cost pressure. This is being seen across TV, IT and

smartphone categories. Second, they are launching higher-specification products to increase perceived consumer value and offset price increases. Several OEMs have already raised operating expectations for high-spec products, according to BOE.

## Earnings Revisions and Rating Downgrade

Incorporating mgmt. comments on the display sector and its innovation business guidance, we raise our 2026/2027 net profit estimates by 9% and 6%, mainly on higher innovation business revenue growth partly offsetting consumer electronics weakness. We also introduce 2028 estimates. We derive a new target price of Rmb8.7 (from Rmb5.0), based on 2.3x P/B based on 2026 BVPS (previous: 1.4x 2025E P/B), set at its 5-year multiple peak to reflect concentration and profitability improvement in the LCD business and likely positive market sentiment on glass substrate applications starting from 2027.

We believe BOE could be a key beneficiary once glass substrate is widely adopted in China, but adoption timeline and penetration still look unclear. Meanwhile, as BOE anticipates it will reach a definitive mass-production investment decision by mid-2027, and production line construction would likely take 2 years, we believe glass substrate is unlikely to generate meaningful net profit contribution before 2029. We downgrade BOE from Buy to Neutral as we believe current valuation has factored in the improving LCD business and potential glass substrate contribution.

Figure 1. BOE estimate changes

| Rmb m               | 2026E    |          |         | 2027E    |          |         | 2028E    |      |      |
|---------------------|----------|----------|---------|----------|----------|---------|----------|------|------|
|                     | New      | Old      | Chg     | New      | Old      | Chg     | New      | Old  | Chg  |
| Revenue             | 202,658  | 213,703  | -5%     | 212,084  | 225,488  | -6%     | 227,114  | n.a. | n.a. |
| Gross profit        | 32,951   | 33,254   | -1%     | 36,096   | 36,170   | 0%      | 40,867   | n.a. | n.a. |
| OPEX                | (22,307) | (21,992) | 1%      | (23,407) | (22,736) | 3%      | (25,005) | n.a. | n.a. |
| Operating profit    | 10,644   | 11,262   | -5%     | 12,689   | 13,434   | -6%     | 15,861   | n.a. | n.a. |
| Pre-tax profit      | 9,781    | 9,630    | 2%      | 11,836   | 12,427   | -5%     | 15,047   | n.a. | n.a. |
| Net profit          | 7,560    | 6,933    | 9%      | 9,469    | 8,947    | 6%      | 12,037   | n.a. | n.a. |
| Ratio (%)           |          |          |         |          |          |         |          |      |      |
| Gross margin        | 16.3%    | 15.6%    | 0.7ppts | 17.0%    | 16.0%    | 1.0ppts | 18.0%    | n.a. | n.a. |
| OPEX to Sales ratio | 11.0%    | 10.3%    | 0.7ppts | 11.0%    | 10.1%    | 1.0ppts | 11.0%    | n.a. | n.a. |
| Operating margin    | 5.3%     | 5.3%     | 0.0ppts | 6.0%     | 6.0%     | 0.0ppts | 7.0%     | n.a. | n.a. |
| Net margin          | 3.7%     | 3.2%     | 0.5ppts | 4.5%     | 4.0%     | 0.5ppts | 5.3%     | n.a. | n.a. |

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Source: Citi Research

## Bull/Bear: BOE Technology (000725.SZ)

Rmb **11.400**



▲ 50% Upside

Rmb **8.700**



▲ 15% Upside

Rmb **4.600**



▼ 39% Downside

Rmb



Spread 90pp  
Current Price and expected returns (upside/downside) as of 07 Jul 2026



### BULL Assumptions

- PB at 2.3x on panel biz and PE of 50x on potential glass substrate contribution



### BASE Assumptions

- PB at 2.30x



### BEAR Assumptions

- PB at 1.1x, the 5-year average



## BOE Technology

### Company description

Founded in 1993, BOE was listed on the Shenzhen Stock Exchange in 1997. It focuses on manufacturing both large-sized and mid-to-small-sized panels. BOE's TFT-LCD production lines include Gen 4, Gen 5, Gen 6, and Gen 8. It also has invested in a Gen 10.5 fab in Hefei. According to IHS, BOE's Gen 10.5 fab started producing 65-inch panels and shipped to several TV makers in April 2018.

### Investment strategy

We have a Neutral rating on BOE as we believe current valuation has factored in below fundamentals: 1) LCD S/D improving, driving BOE's fundamental improvement and ROE increase; 2) BOE is more resilient than peers on leadership, cost efficiency, and diversified portfolio, and its risk/reward profile looks attractive; 3) BOE should be a key beneficiary of glass substrate adoption in China.

### Valuation

Our target price of Rmb8.7 is based on 2.3x our 2026E book value per share estimate, set at its 5-year multiple peak to reflect concentration and profitability improvement in the LCD business and likely positive market sentiment on glass substrate applications starting from 2027. We use P/B to value BOE as it is a cyclical, asset-heavy company, making P/B multiple an appropriate metric, in our view.

### Risks

Key risks that could lead the share price to deviate from our target price include: (1) faster/slower-than-expected panel price recovery; (2) stronger/weaker-than-expected panel demand; (3) higher/lower-than-expected sell-through; and (4) accelerated/decelerated glass substrate adoption or applications.

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## Appendix A-1

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## IMPORTANT DISCLOSURES

### BOE Technology (000725.SZ)

#### Ratings and Target Price History Fundamental Research

Analyst: Karen Huang



|   | Date               | Rating | Target Price | Closing Price |
|---|--------------------|--------|--------------|---------------|
| 1 | 31-Aug-23 11:48:52 | 1H     | *5.20        | 4.01          |
| 2 | 01-Nov-23 11:19:28 | 1H     | *4.70        | 3.75          |
| 3 | 02-Apr-24 18:42:12 | *1     | *5.20        | 4.30          |

|   | Date               | Rating | Target Price | Closing Price |
|---|--------------------|--------|--------------|---------------|
| 4 | 29-Apr-24 13:01:42 | 1      | *5.10        | 4.34          |
| 5 | 28-Aug-24 10:55:19 | 1      | *4.60        | 3.82          |
| 6 | 07-Oct-24 13:31:27 | 1      | *5.30        | 4.47          |

|   | Date               | Rating | Target Price | Closing Price |
|---|--------------------|--------|--------------|---------------|
| 7 | 29-Apr-25 03:05:39 | 1      | *5.10        | 3.82          |
| 8 | 31-Oct-25 05:07:10 | 1      | *5.00        | 4.06          |

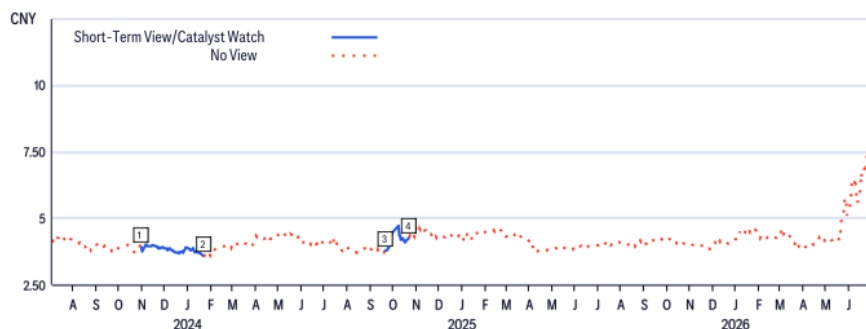
\*Indicates Change

Rating/target price changes above reflect Eastern Time

### BOE Technology (000725.SZ)

#### Short-Term View/Catalyst Watch Research

Analyst: Karen Huang



|   | Date               | Action    | Expected Direction | Duration | Closing Price |
|---|--------------------|-----------|--------------------|----------|---------------|
| 1 | 31-Oct-23 04:06:28 | Add CW    | Downside           | 90 Days  | 3.92          |
| 2 | 22-Jan-24 15:25:10 | Remove CW | Downside           | 90 Days  | 3.58          |

|   | Date               | Action    | Expected Direction | Duration | Closing Price |
|---|--------------------|-----------|--------------------|----------|---------------|
| 3 | 22-Sep-24 14:07:10 | Add CW    | Upside             | 30 Days  | 3.78          |
| 4 | 22-Oct-24 23:28:37 | Remove CW | Upside             | 30 Days  | 4.28          |

CW - Catalyst Watch, STV - Short-Term View

Rating/target price changes above reflect Eastern Time

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| Data current as of 01 Jul 2026                                             | 12 Month Rating |      |      | Catalyst Watch |      |      |
|----------------------------------------------------------------------------|-----------------|------|------|----------------|------|------|
|                                                                            | Buy             | Hold | Sell | Buy            | Hold | Sell |
| Citi Research Global Fundamental Coverage (Neutral=Hold)                   | 61%             | 32%  | 7%   | 36%            | 48%  | 16%  |
| % of companies in each rating category that are investment banking clients | 38%             | 42%  | 27%  | 42%            | 37%  | 36%  |

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